Business Performance Analysis

Sales, Profit and Discount Review

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Tools used	Microsoft Power BI (Power Query and DAX) and Microsoft Excel
Status	For review

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Executive summary

The three points a manager needs from this report

01

Two segments bring 80% of sales

Government and Small Business together give 80% of all sales. Government alone gives 67% of the profit, so it is the segment the business can least afford to lose.

02

Enterprise sells a lot but loses money

Enterprise brings $19.6M in sales but a loss of $0.61M. Every Enterprise order with a discount of 5% or more lost money. Every order at 4% or less made money.

03

High discounts cut the margin in half

Orders with a 10% to 15% discount keep about 9 cents of profit from each dollar of sales. Orders with a 1% to 4% discount keep about 18 cents.

Note: the old year on year cards showed 349% because 2013 has only four months of data. They have been removed. Like for like (Sep to Dec), sales grew 37% and profit 40%. See page 13.

Scope and data

What I looked at, and the questions I started with

01

Where do sales and profit come from?

Which segments, countries and products matter most

02

Is any part of the business losing money?

And what is causing it

03

What do discounts do to profit?

Whether bigger discounts pay for themselves

The dataset

700

sales records

$118.7M

sales after discounts

$16.9M

profit

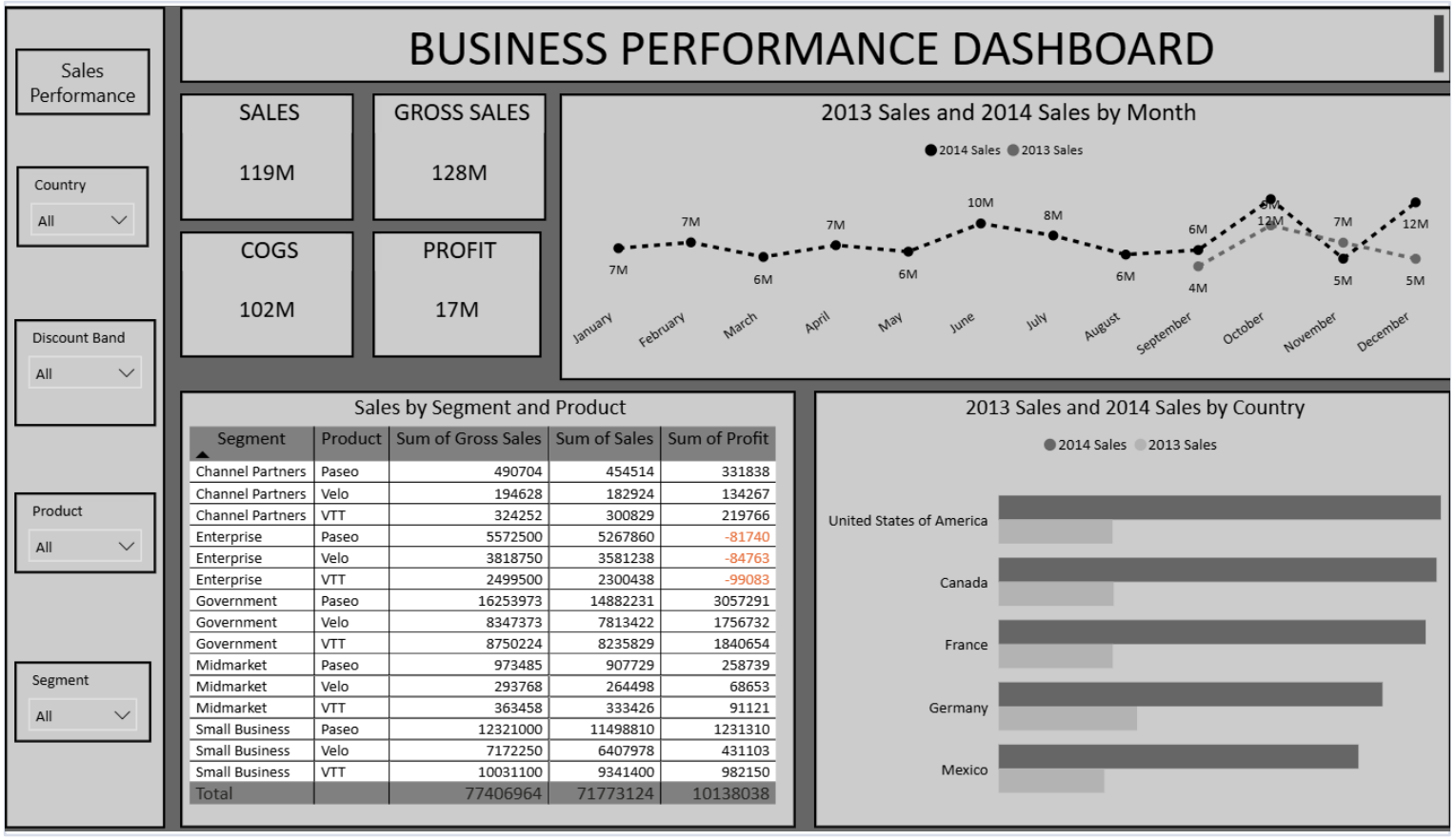
Sep 2013 to Dec 2014

5 segments, 5 countries, 6 products

Method: the data was loaded into Power BI with Power Query, set up as a star schema, and measured with DAX. Every total in this report was checked back against the Excel source file.

The Power BI dashboard

One report page, with filters that control every visual



Report page built in Power BI Desktop on the public Financial Sample dataset.

What the page shows

KPI cards

Sales, gross sales, cost of goods sold and profit for the selected filters.

Filters (slicers)

Country, discount band, product and segment. Every visual reacts to them.

Monthly trend

Sales by month, 2014 against 2013.

Segment table

Products in each segment with sales and profit. Losses show in orange.

Country bars

2013 and 2014 sales side by side for each country.

How I fetched and prepared the data

The ETL flow, from source file to a dashboard that refreshes itself

1

Fetch

Get Data in Power BI. Here an Excel file. At work: SQL Server, SharePoint, CSV exports or APIs.

2

Clean

Power Query fixes types, names and values. Every step is saved and runs again on refresh.

3

Model

One sales table in the middle, with date, product, segment and country around it.

4

Measure

DAX measures for sales, profit, margin and year on year.

5

Share

Publish to Power BI Service. Scheduled refresh through a data gateway.

In plain words

Fetch means pulling the data in. Clean means making every row follow the same rules. Model means linking the tables so filters work. I keep all cleaning inside Power Query, so when new data arrives the same steps run again without anyone touching them.

When the data sits in a database

I write the SQL query or view first, so only the columns and rows the report needs come in. Heavy joins and filters run in the database, not in Power BI. The report then refreshes on a schedule through the gateway, and large tables use incremental refresh.

Cleaning steps and checks

What I changed in Power Query, and why

#	Step	What I did	Why it matters
1	**Set the right data types**	Date as a date. Money and units as decimal numbers, because some rows have half units.	Totals and date filters only work with the right types.
2	**Fix column names**	The Sales column had a hidden space at the start. Renamed it.	Hidden spaces break measures and are hard to spot.
3	**Make values readable**	Changed the discount band value "None" to "No discount".	Nobody mistakes it for missing data on a slicer.
4	**Remove repeat columns**	Month number, month name and year all come from the Date column. Moved them into a date table.	One source for time, so dates never disagree.
5	**Add a date table**	A calendar from Sep 2013 to Dec 2014, marked as the date table.	Needed for correct year on year measures.
6	**Check the maths on every row**	Gross sales minus discounts equals sales. Sales minus COGS equals profit.	All 700 rows matched, so the totals can be trusted.

Data model and measures

A star schema: one sales table in the middle, lookup tables around it



Key measures (DAX)

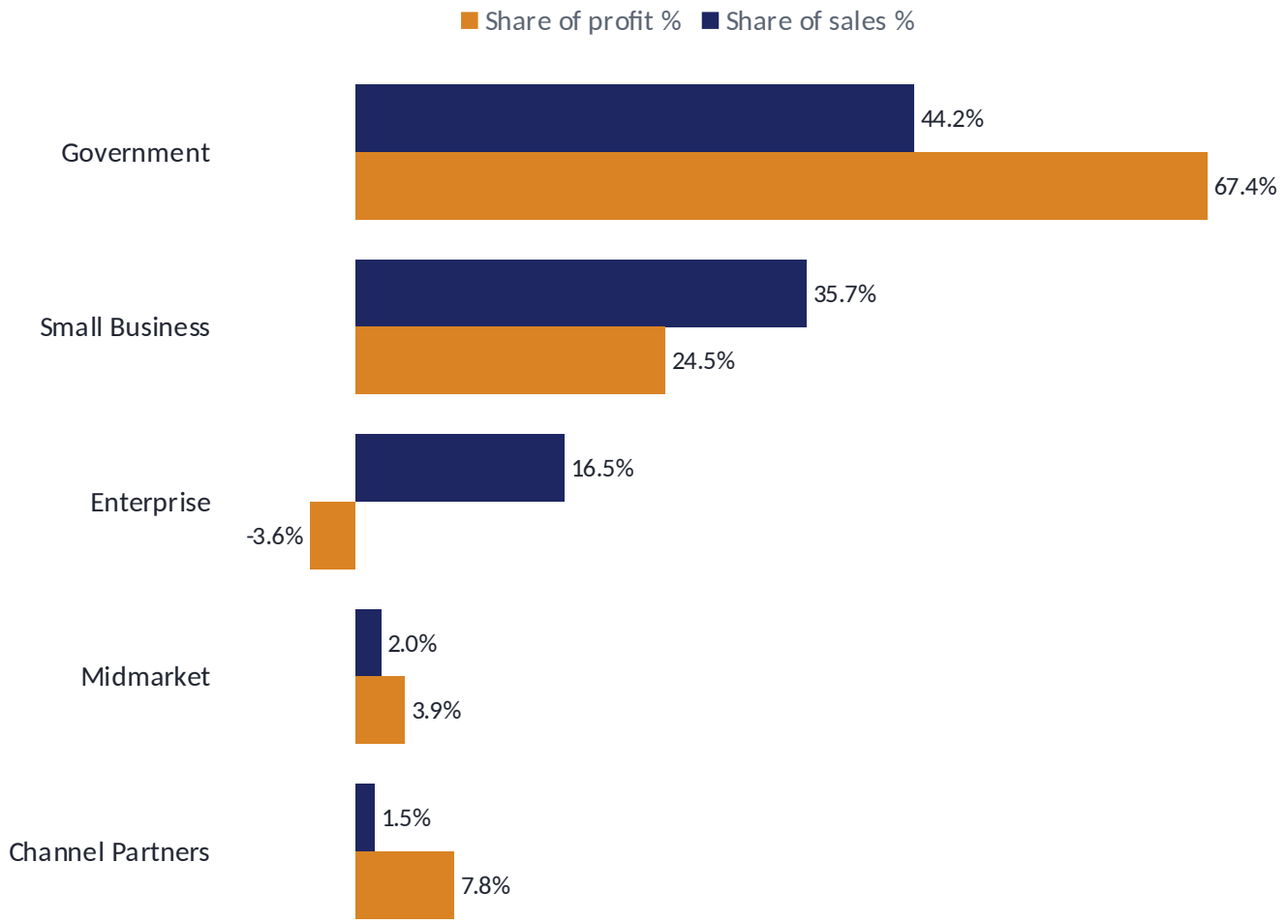
```
Total Sales = SUM ( Sales[Sales] )
Total Profit = SUM ( Sales[Profit] )
Profit Margin % =
    DIVIDE ( [Total Profit], [Total Sales] )
Sales Last Year =
    CALCULATE ( [Total Sales],
        SAMEPERIODLASTYEAR ( 'Date'[Date] ) )
YoY % = DIVIDE ( [Total Sales] - [Sales Last Year],
    [Sales Last Year] )
```

In plain words

A measure is a saved calculation. It works out the answer again every time someone clicks a filter. The lookup tables hold each product, date and country once, so a filter on one of them flows to the sales table.

Two segments give 80% of sales

Share of total sales and total profit by customer segment



80%

of all sales come from Government and Small Business

What this means

- Government gives 67% of profit from 44% of sales. It is the most important segment.
- Channel Partners is small, only 1.5% of sales, but keeps 73 cents of profit from every dollar.
- Enterprise gives 16.5% of sales and takes profit away. The next page shows why.

Enterprise sells a lot but loses money

Every Enterprise order with a discount of 5% or more made a loss

$19.6M

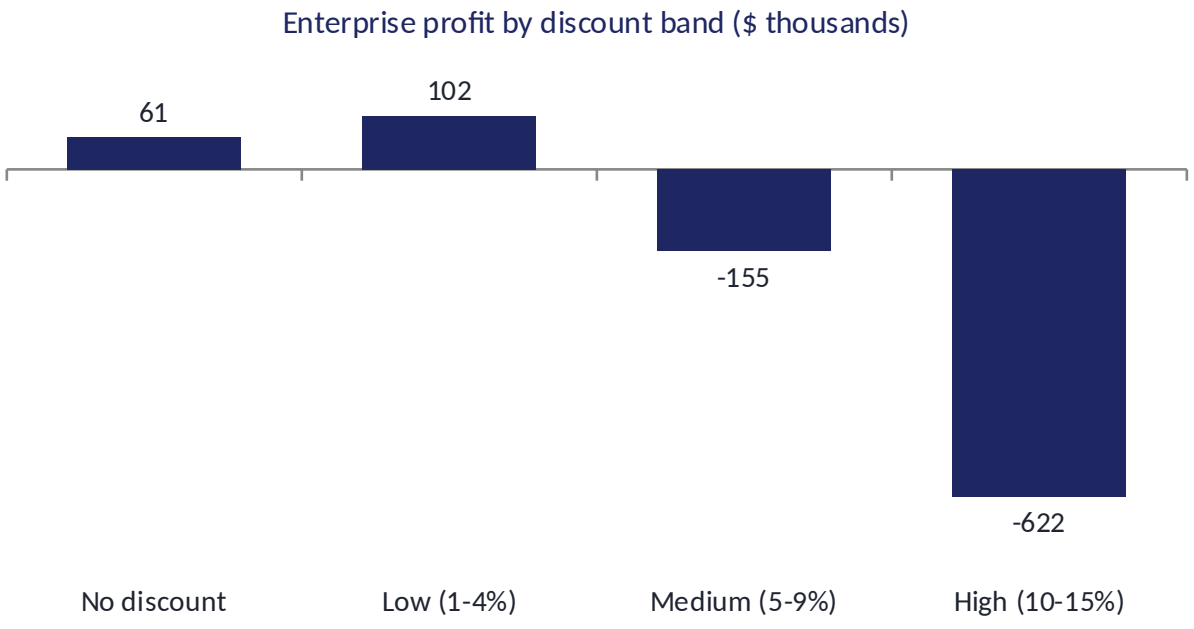
Enterprise sales

-$0.61M

Enterprise profit

58 of 100

Enterprise orders made a loss



Why it happens

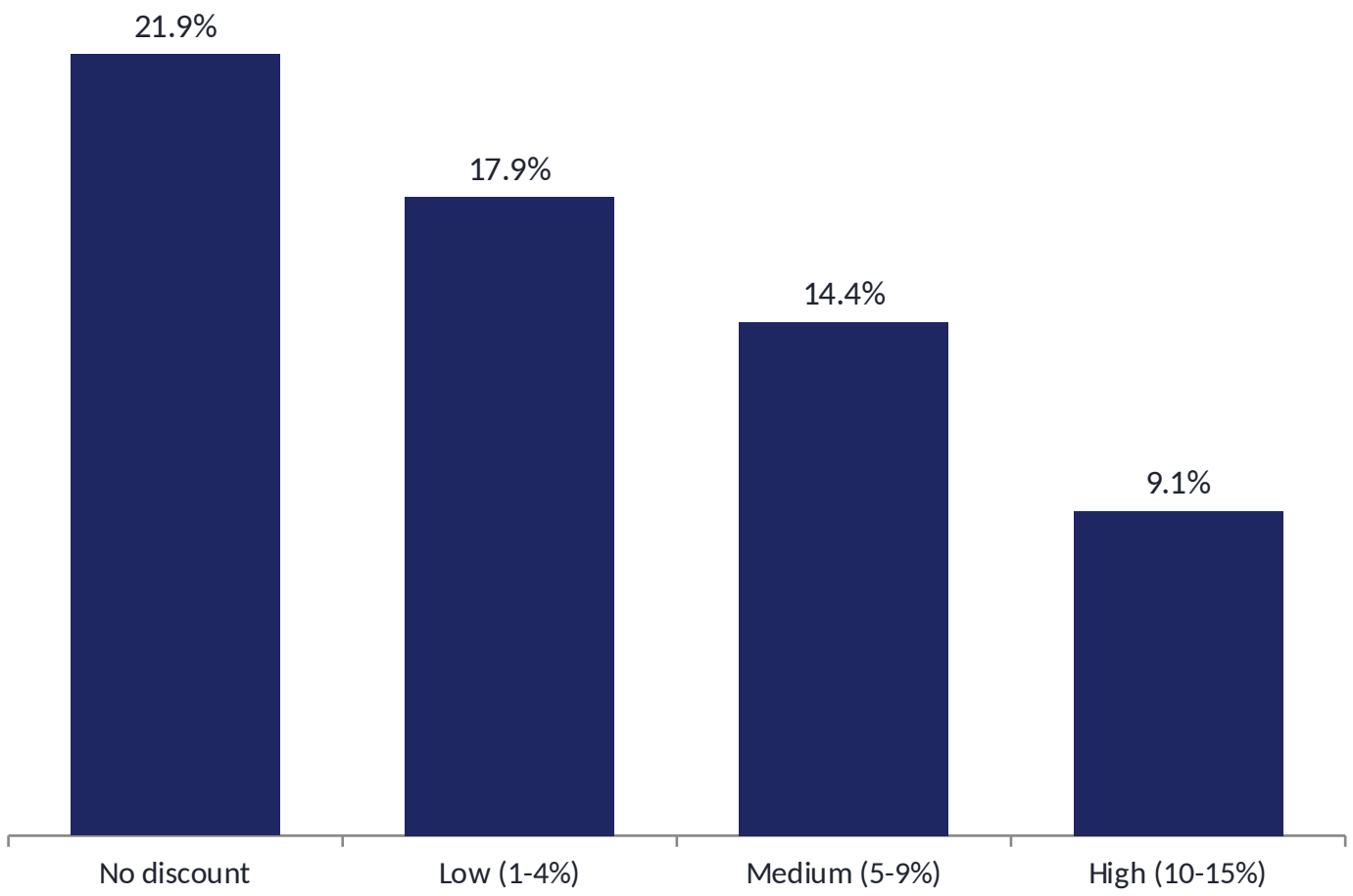
Enterprise units sell for $125 and cost $120. That leaves $5 of room, which is 4% of the price. Any discount bigger than 4% turns the order into a loss. The data follows this rule exactly.

The question it raises

Who approves Enterprise discounts above 4%, and do they know each one loses money?

High discounts cut the margin in half

Profit margin by discount band, all segments



31% vs 20%

High discount orders give 31% of sales but only 20% of profit

What this means

- Every step up in discount takes margin away.
- Orders with no discount keep 22 cents per dollar. High discount orders keep 9.
- A discount only pays if it brings enough extra volume. This data does not show that it does.

Country is not a strong split

Sales, profit and margin by country, both years together

Rank	Country	Sales	Share of sales	Profit	Margin
1	**United States**	$25.03M	21.1%	$3.00M	12.0%
2	**Canada**	$24.89M	21.0%	$3.53M	14.2%
3	**France**	$24.35M	20.5%	$3.78M	15.5%
4	**Germany**	$23.51M	19.8%	$3.68M	15.7%
5	**Mexico**	$20.95M	17.6%	$2.91M	13.9%
	Total	**$118.73M**	**100.0%**	**$16.89M**	**14.2%**

Source: Financial Sample dataset, 700 records. Figures rounded.

Reading the table

Every country sits between 18% and 21% of sales. The United States sells the most but keeps the least profit per dollar.

Country on its own explains little. Segment and discount explain much more, so that is where attention should go.

Data quality and dashboard notes

Four things to fix or confirm before version 2

1

The old year on year cards compared 12 months with 4

2013 only has September to December, so the old cards showed 349%. Comparing the same four months, sales grew 37% and profit grew 40%. The cards have been removed from the dashboard.

2

The top 3 table shows the same three products for every segment

The table takes one top 3 for the whole business, so every segment shows Paseo, Velo and VTT. For Channel Partners the real top 3 are Paseo, Amarilla and VTT. Version 2 ranks products inside each segment.

3

June, October and December 2014 have twice as many records

These months have 70 rows instead of 35. The peaks on the monthly chart come from more records, not bigger orders. To confirm with the data owner.

4

Manufacturing price does not explain COGS

In 536 of 700 rows, units times manufacturing price is not equal to COGS. Cost follows the segment instead. I used COGS as given and would confirm the cost rule with finance.

Sales, discount and profit totals reconcile to the Excel source on all 700 rows.

Recommendations

Four actions, in the order I would take them

01

Cap Enterprise discounts at 4%

Above 4%, every Enterprise order lost money. Either cap the discount or review the Enterprise price.

Owner

Sales and Pricing

02

Add approval for discounts of 10% and above

High discounts bring 31% of sales but only 20% of profit. One approval step would protect margin.

Owner

Finance

03

Protect Government, and test growing Channel Partners

Government gives two thirds of profit. Channel Partners keeps a 73% margin on a small base, so a small test is worth it.

Owner

Sales

04

Fix the dashboard before the next review

Like for like year on year (Sep to Dec only), a top 3 table ranked inside each segment, and confirmed rules for cost and record counts.

Owner

Reporting

Thank you

Happy to walk through the Power BI file, or to build the same kind of analysis on your own data.

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